Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration*	**Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky.
12/03/26

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Session	**2025-26 (EVEN)**	**Course Name**	**JAVA FSD Lab**
Semester	**6**	**Course Code**	**23IOT1607**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**06**
Course Outcome	1. Apply OOP concepts to solve the real-life problem. 2. Use the database concepts to integrate in giving OOP oriented solution. 3. Develop full-stack applications integrating frontend, backend, and database with security. 4. Design and develop mini-project with frontend, backend, and database integration, including authentication and testing.
Aim	Apply the features of stream API to implement the solution
Problem Definition	**Classroom Performance Tracker** A teacher has a list of student marks. The teacher needs to quickly: Filter students who passed (scored 40 or above). Add Grace Marks of 5 points to every student. Count how many students scored distinction (above 75) after grace marks.
Theory (100 words)	The Java Stream API is very useful for implementing the Classroom Performance Tracker because it allows the teacher to process the list of student marks in a simple, readable, and efficient way without writing multiple loops. In this problem, the filter() method is used to identify the students who have passed by selecting only those marks that are 40 or above. Next, the map() method is applied to add 5 grace marks to every student's score, which transforms each mark into a new updated value. After adding grace marks, the program again uses filter() to check which students scored more than 75, meaning they achieved distinction. Finally, the count() method is used to calculate the total number of distinction students. Thus, by using Stream API features like filter, map, and count, the teacher can easily perform all required operations in a clean and modern way. This approach reduces code complexity, improves readability, and makes data

	processing faster and more organized.
Procedure and Execution (100 Words)	1. Start the program and import required packages (`java.util` and `java.util.stream`). 2. Create a `List<Integer>` to store student marks. 3. Convert the list into a stream using `stream()`. 4. Use `filter()` to select students with marks ≥ 40. 5. Use `map()` to add 5 grace marks. 6. Use `forEach()` or `count()` as terminal operations. 7. End the program.

	import java.util.*;
	import java.util.stream.*;
	
	public class Main {
	public static void main(String[] args) {
	
	// Step 1: Create list of student marks
	List<Integer> marks = Arrays.asList(35, 42, 67, 78, 90, 55, 38, 81);
	
	System.out.println("Final Marks after Grace:");
	
	List<Integer> finalMarks = marks.stream()
	.filter(m -> m >= 40) // passed students
	.map(m -> m + 5) // add grace marks
	.collect(Collectors.toList());
	
	finalMarks.forEach(m -> System.out.println(m));
	
	long distinctionCount = finalMarks.stream()
	.filter(m -> m > 75)
	.count();
	
	System.out.println("Number of students with distinction: " + distinctionCount);
	}}

	Output: PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD> c:; cd 'c:\Users\sheik\OneDrive\Desktop\JAVAFSD' ; & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sheik\AppData\Roaming\Code\User\workspaceStorage\d7201e80c8ec46414256147816d4f70a\redhat.java\jdt_ws\JAVAFSD_b4be34d3\bin' 'Main' Final Marks after Grace: 47 72 83 95 60 86 Number of students with distinction: 3 PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>
Output Analysis	The output shows the final marks after adding 5 grace marks to each student using the Stream API map() method. The updated marks displayed are 47, 72, 83, 95, 60, and 86, which means each original mark has been increased by 5. After this, the program checks how many students scored more than 75, which is the condition for distinction. From the final marks, the students who achieved distinction are 83, 95, and 86. Therefore, the total number of students with distinction is 3, which is correctly shown in the output. This output proves that the program successfully applied the Stream API features such as map() for adding grace marks and filter() + count() for identifying and counting distinction students. The program is working correctly and gives the expected result.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	In conclusion, the Classroom Performance Tracker successfully demonstrates the use of Java Stream API to process student marks in a simple and efficient way. By using filter(), the program can identify passed students, with map() it adds 5 grace marks to every student, and with count() it calculates the number of students who achieved distinction

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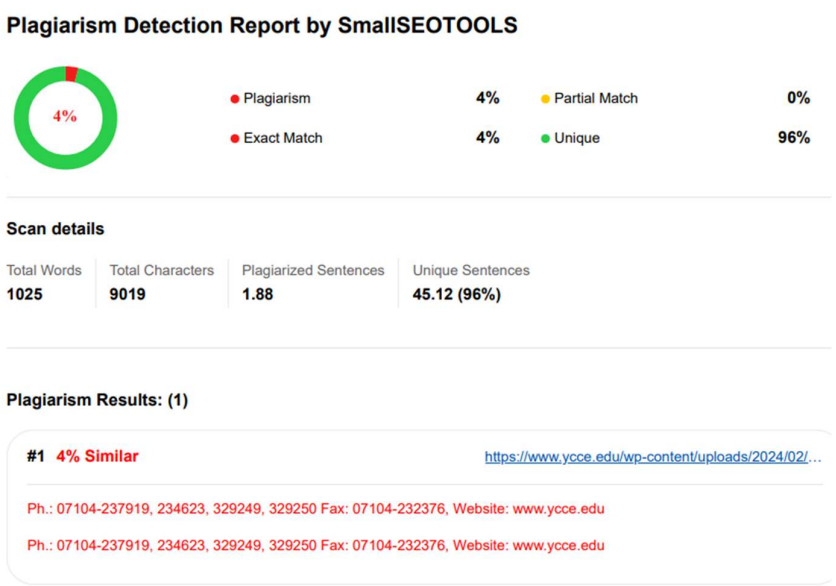
	after grace marks. The output confirms that the program works correctly, as the final updated marks are displayed and 3 students scored above 75. Thus, the Stream API makes the solution more readable, concise, and effective compared to traditional looping methods.
Plag Report (Similarity index < 12%)	
Date	**12/03/2026**

Paste here Faculty signed observation page.

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PRACTICAL No:- 06

Aim:- Apply the features of stream API to implement the solution.

problem defination : classroom performance Tracker.

A Teacher has a list of student marks. The teacher needs to quickly: filter student who passed (scored 40 or above). Add grace mark of 5 points to every student. Count how many student scored distinction (above 75) after grace marks.

output:- final Marks after Grace:
47
72
83
95
60
86

Number of student with distinction:

12/3/26

Shrikrupa